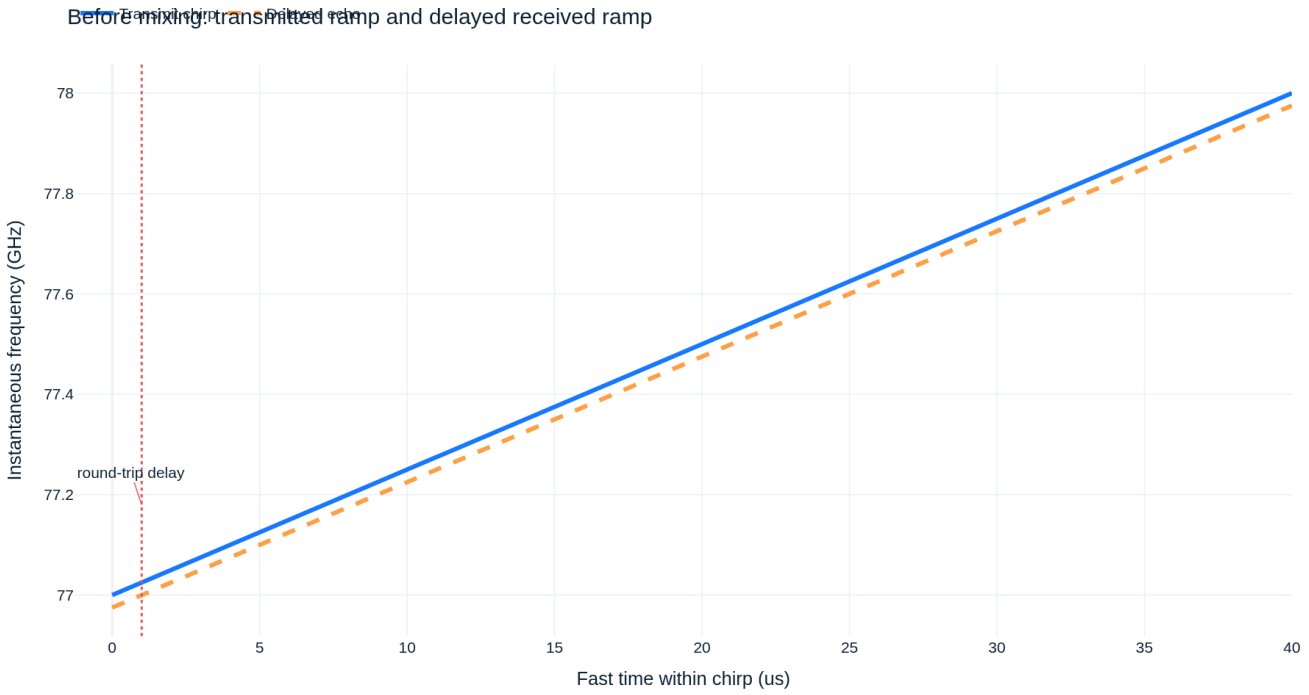
FMCW Radar Data Evolution and Research Code Architecture

A visual, stage-by-stage guide from transmitted chirp to KPI validation



Purpose. Show what the data looks like before and after every major radar-processing stage, where each research module fits, what mathematics changes the representation, and how to validate the result. Numerical plots combine documented research results with clearly labeled synthetic examples.

Prepared from the radar monograph, research ideas ledger, and executed experiment storage ledger. October 2026.

How to read this guide

Each section uses the same four questions: What enters? What mathematical operation is applied? What does the output look like? What can fail? “Before and after” plots are included so the transformation is visible rather than only described.

Part	Pages	What you will learn
A	3-5	System map, data dimensions, code and folder architecture
B	6-13	RF chirp through range-Doppler and CFAR
C	14-18	CDC, beamforming, point clouds, RDU
D	19-23	Tracking, gating, multi-radar fusion, CAN, KPI
E	24-26	HPCC, research experiment design, integration checklist

Notation used throughout

Symbol	Meaning	Typical unit
t	fast time inside one chirp	s
m	slow-time chirp index	count
k, l	range and Doppler bins	index
r, v_r	range and radial velocity	m, m/s
theta, phi	azimuth and elevation	rad or deg
P, R	track and measurement covariance	state-unit squared
P_fa	false-alarm probability	dimensionless

1. Full processing map

The representation changes at every step. A continuous electromagnetic waveform becomes complex ADC samples, spectra, sparse cells, physical detections, state estimates, serialized integers, and finally KPI scores.



2. What shape is the data at each stage?

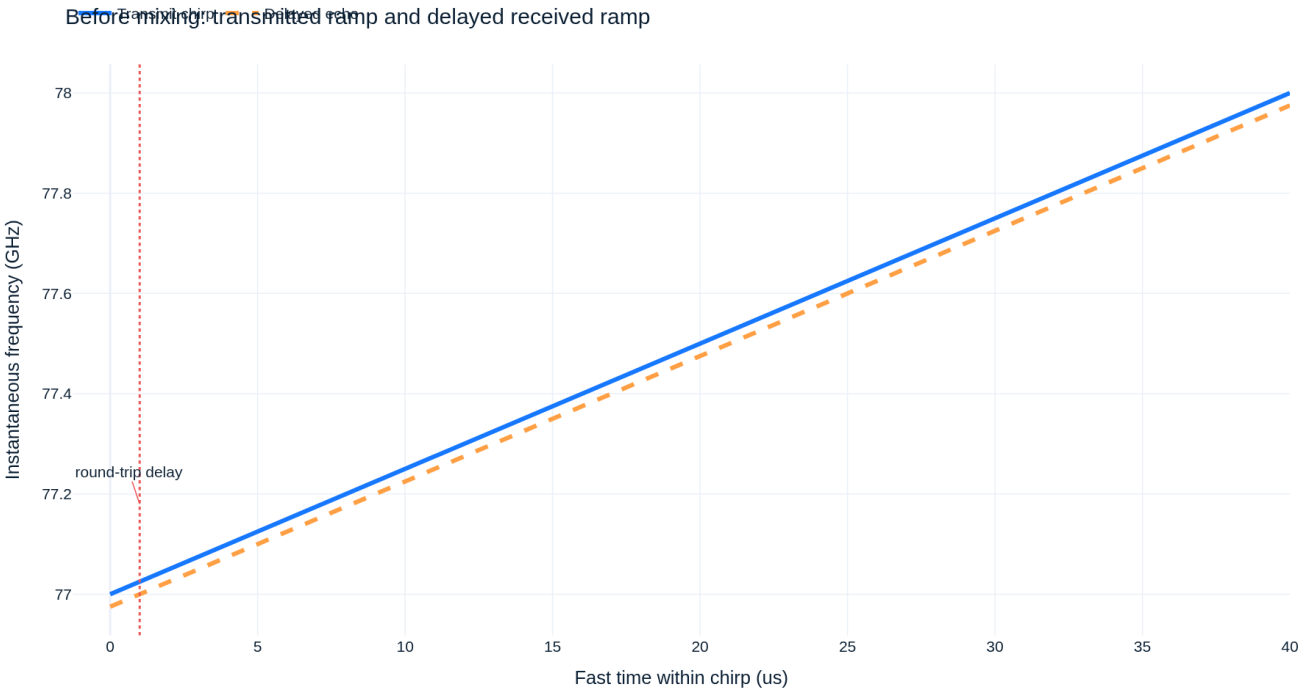
Stage	Representative shape	Typical content
I/Q ADC	[chirp, sample, RX]	complex voltage samples
Range FFT	[chirp, range_bin, RX]	complex beat spectrum
Doppler FFT	[range_bin, Doppler_bin, RX]	complex range-Doppler spectrum
Power map	[range_bin, Doppler_bin]	non-coherent energy
CDC	list of records	bin indices, energy, beamvector
AF detections	list of detections	r, v_r, az, el, RCS, SNR, flags
Tracks	list of states	x, y, vx, vy, covariance, ID
Fusion output	global track list	common epoch and common frame
CAN-FD	byte payloads	quantized fields through DBC
KPI	matched pairs and summaries	yield, accuracy, latency, pass/fail

Research folder architecture

resim_research/ contains isolated candidate modules. edge_hdf/ contains sensor HDF inputs. mf4_data/ contains manifests and recording paths. hdf5_to_kpi_csv.py and candidate_mf4.py bridge candidate outputs into the validation flow. storage.md stores the executed evidence, failure analysis, and verdict for every run.

Recommended rule: keep the production implementation read-only during the first experiment. A candidate module should expose a small deterministic core function plus a demo or test harness with known truth and asserts.

3. RF chirp: what leaves the radar and what comes back?

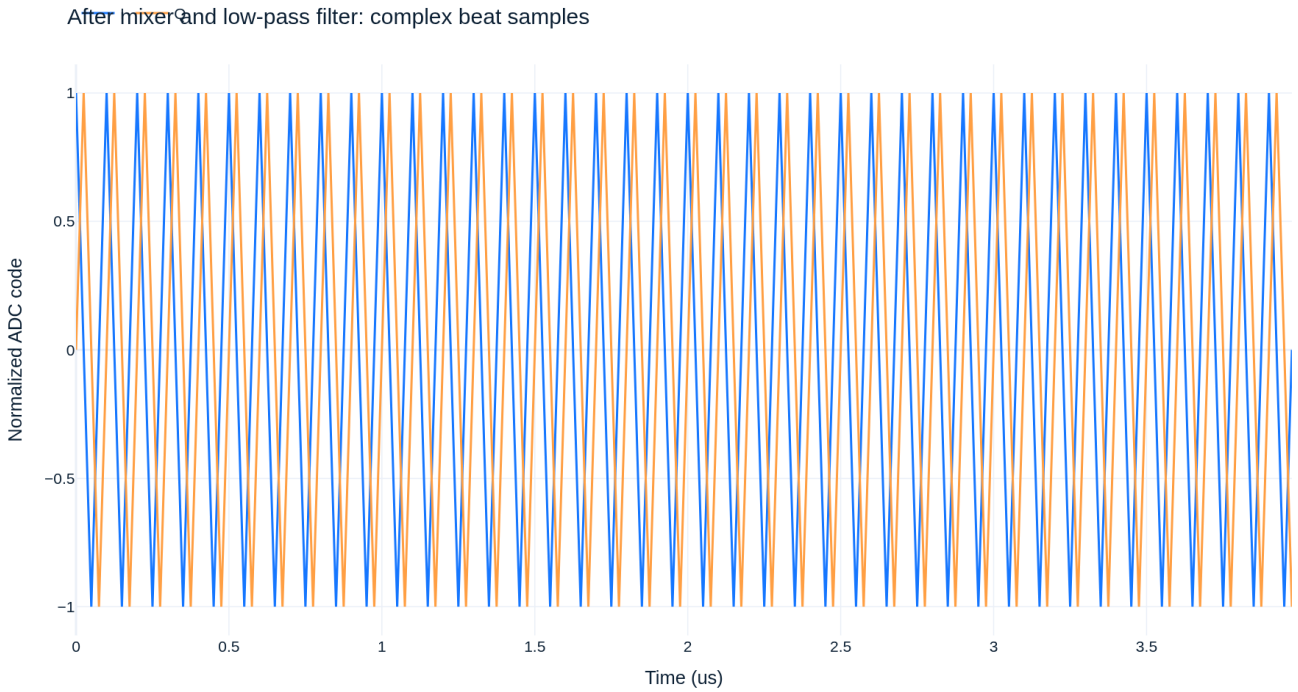


Input. A voltage-controlled oscillator creates an upward linear frequency sweep from f_c to f_c+B over T_chirp. Scene interaction. The target returns a delayed, weaker copy. The delay is tau=2r/c because the wave travels out and back. Output. Two ramps with nearly identical slope but a small vertical frequency separation.

Core math. S=B/T_chirp and f_b approx S tau = 2Sr/c. At B=1 GHz, T_chirp=40 us, r=150 m, the beat is 25 MHz. Omitting the round trip creates the 12.5 MHz one-way error.

What can fail? RF leakage, target acceleration during the chirp, chirp nonlinearity, interference from another radar, insufficient ADC bandwidth, and incorrect sign conventions.

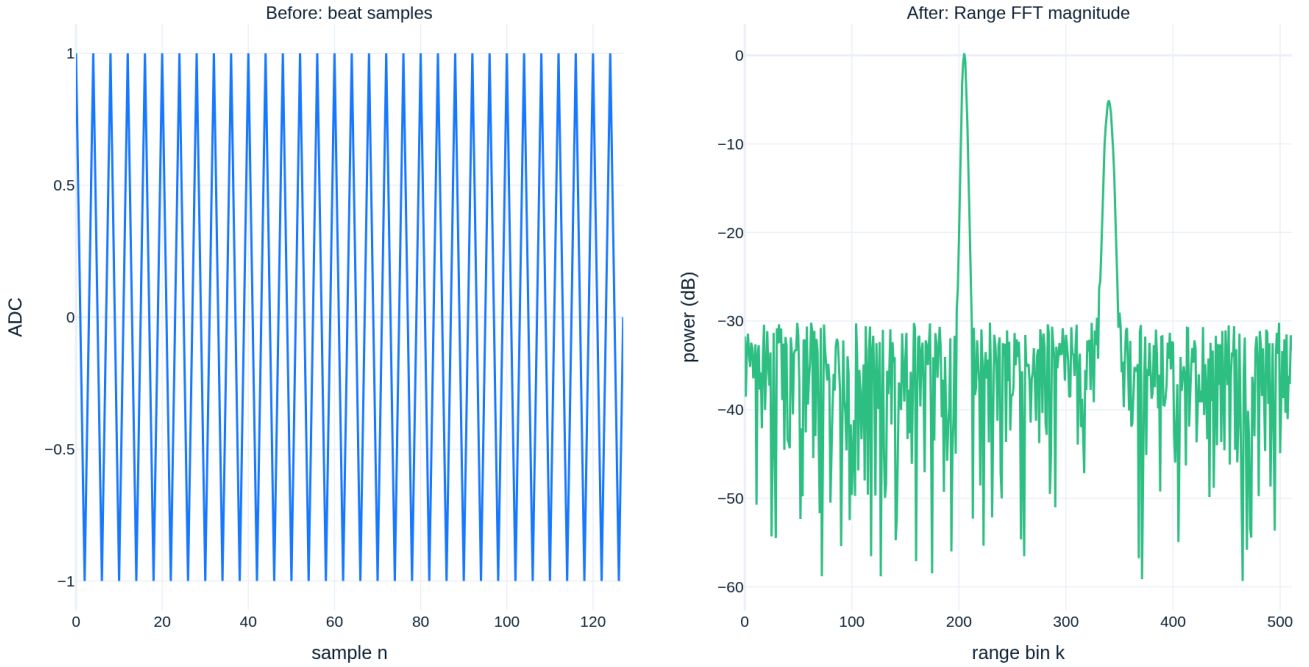
4. Mixer and I/Q ADC: waveform becomes complex numbers



The mixer multiplies the fresh transmit chirp by the delayed echo. Product-to-sum creates one component near twice the carrier and one low-frequency difference component. The low-pass filter removes the high-frequency term. Two 90-degree mixer branches produce I and Q, combined as x[n]=I[n]+jQ[n].

Why complex I/Q matters: a real cosine cannot distinguish +f from -f. Complex rotation preserves the sign of frequency, so approaching and receding motion can be separated. A common I/Q scale must be retained for fixed-point processing; independently rescaling I and Q corrupts phase.

5. Range FFT: fast-time pitch becomes range bins

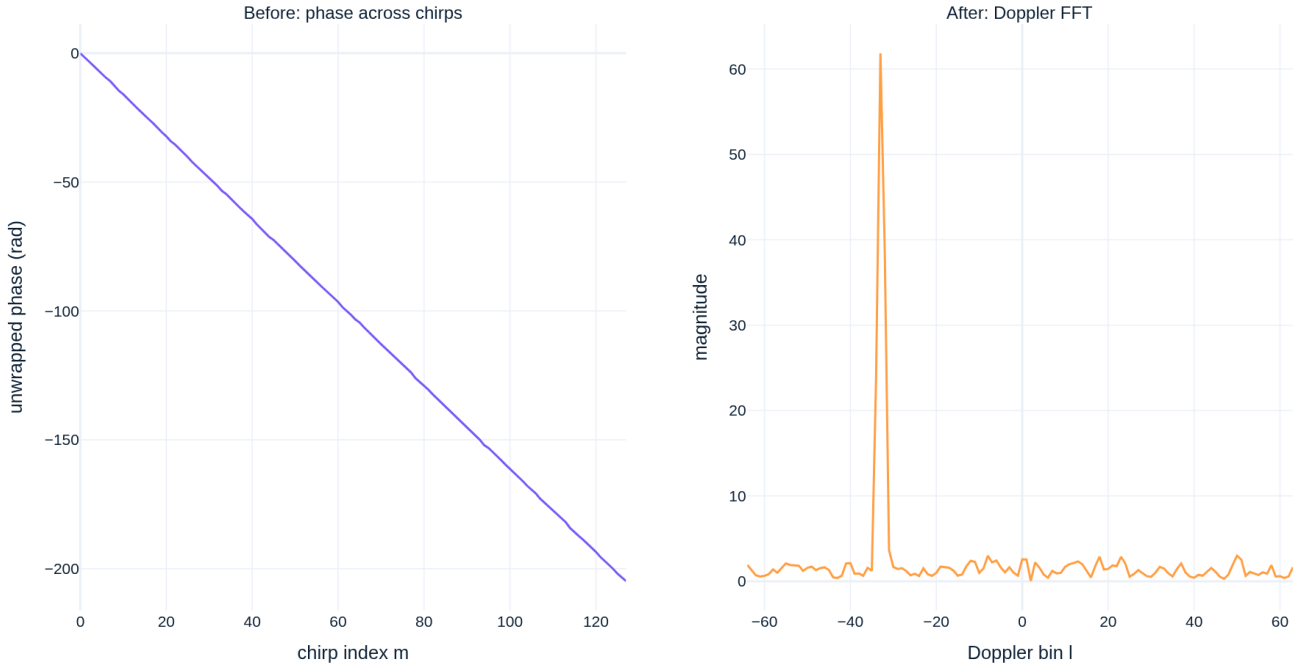


For each chirp, the FFT is taken across ADC samples. A sinusoidal beat concentrates near one frequency bin. The mapping is r_k = k c /(2 S T_s N_R), while true range resolution is approximately c/(2B). Windowing lowers sidelobes but broadens the main lobe.

Before	Operation	After
Complex ADC samples x_m[n]	Window and N_R-point FFT along n	Complex range spectrum X_m[k]
Tone frequency	Bin search or interpolation	Range estimate
Wideband noise	FFT redistributes energy across bins	Noise floor per bin

Code location: an ADC or range-processing experiment can operate before CDC generation. Validate tone-bin mapping, leakage, window loss, saturation, and fixed-point overflow.

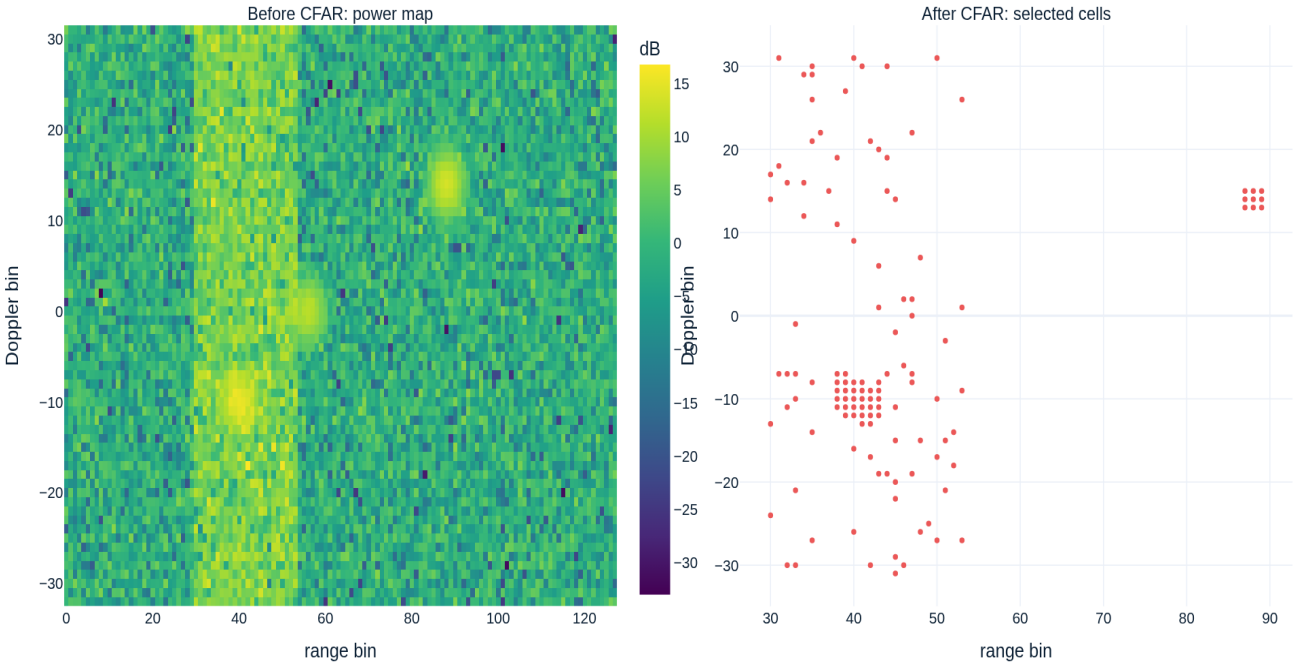
6. Doppler FFT: phase progression becomes radial velocity



At a fixed range bin, the target phase advances across chirps. The slow-time FFT converts this progression into a Doppler bin. With chirp repetition time T_c and N_D chirps, v_l = l lambda /(2 N_D T_c). The unambiguous interval ends near v_max=lambda/(4T_c), so faster targets wrap.

Two-dimensional meaning: range uses samples inside a chirp; Doppler uses the same range bin across many chirps. The output is not yet a list of objects. It is a dense spectral surface where peaks, clutter, and sidelobes coexist.

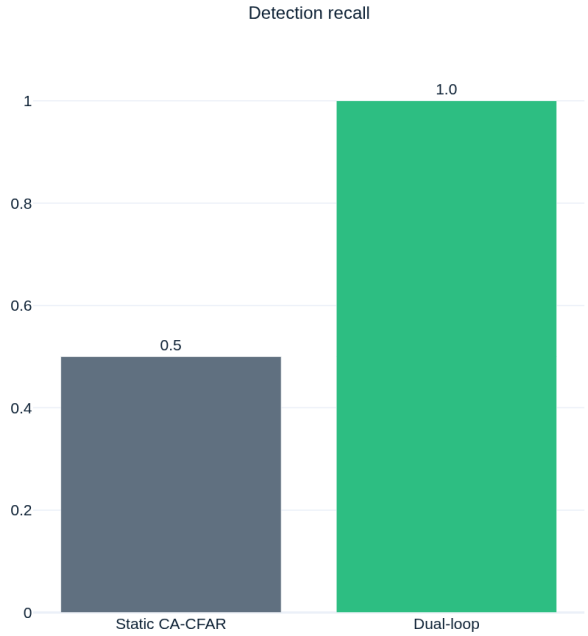
7. Range-Doppler map and CFAR

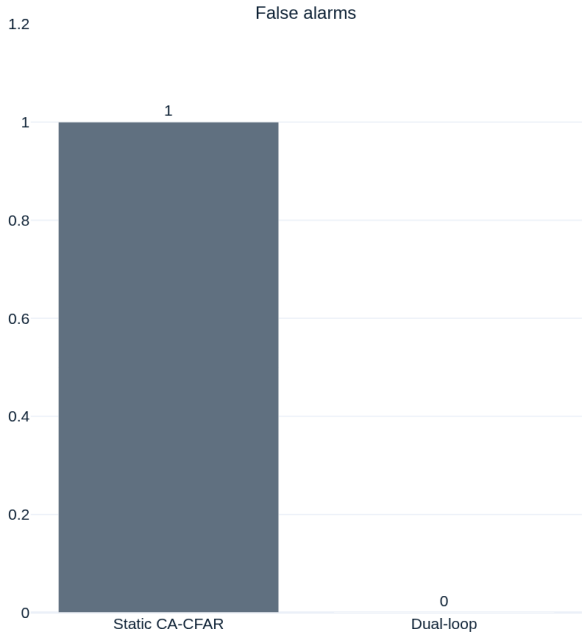


The power map |Y[k,l]|^2 is a two-dimensional image. CFAR estimates local background from reference cells around each cell under test, excludes guard cells, and emits the cells whose power exceeds alpha times the estimated background.

For homogeneous exponential-power clutter, CA-CFAR uses alpha=N(P_fa^(-1/N)-1). A guardrail violates the homogeneous-background assumption, making a target near the wall look weak relative to its neighborhood.

8. CFAR before and after: documented research result



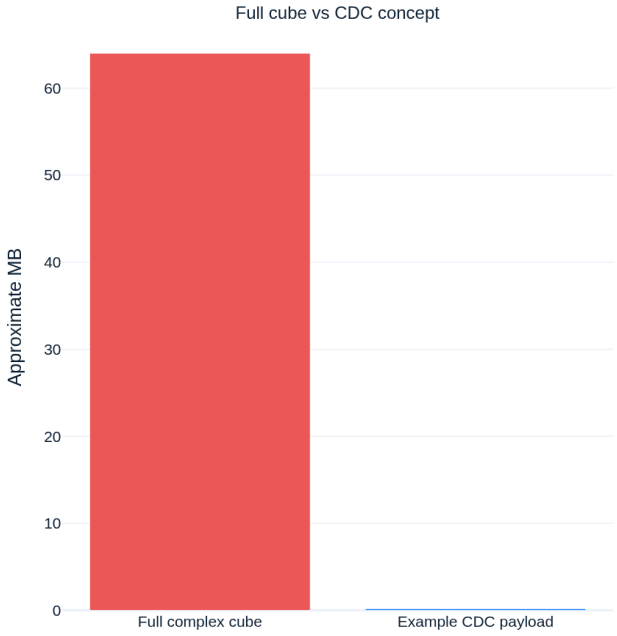


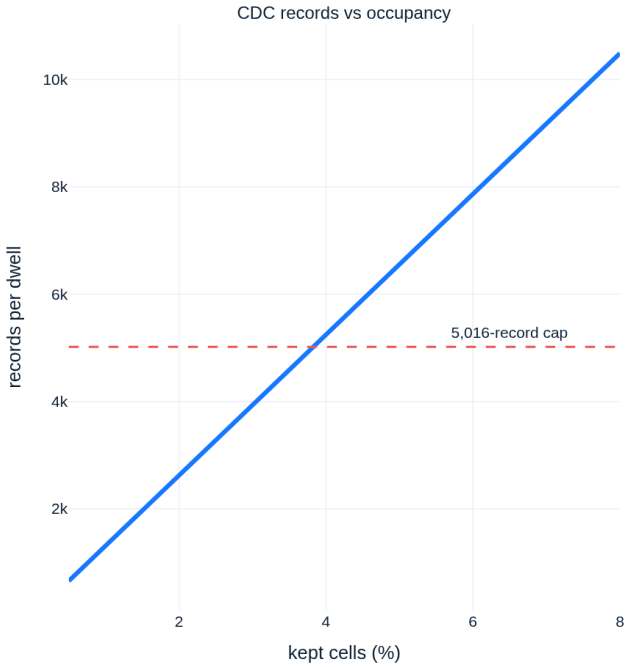
In the executed 128 x 32 synthetic guardrail case, static CA-CFAR achieved 0.50 recall with one false alarm. The dual-loop candidate separated a slow range floor from a fast adaptive window and achieved 1.00 recall with zero false alarms at the same CDC load.

Candidate module: dual_loop_cfar.py. Inputs: power map plus a track-free mask or equivalent prior. Outputs: detection mask, threshold map, and diagnostic state. Tests: edge target, clean target, wall skirt, changing effective N, false-alarm control, and state reset.

Before accepting a detector, report both recall and false alarms. Recall alone can be made perfect by opening the threshold and flooding downstream processing.

9. Compressed Data Cube: dense spectrum becomes sparse records

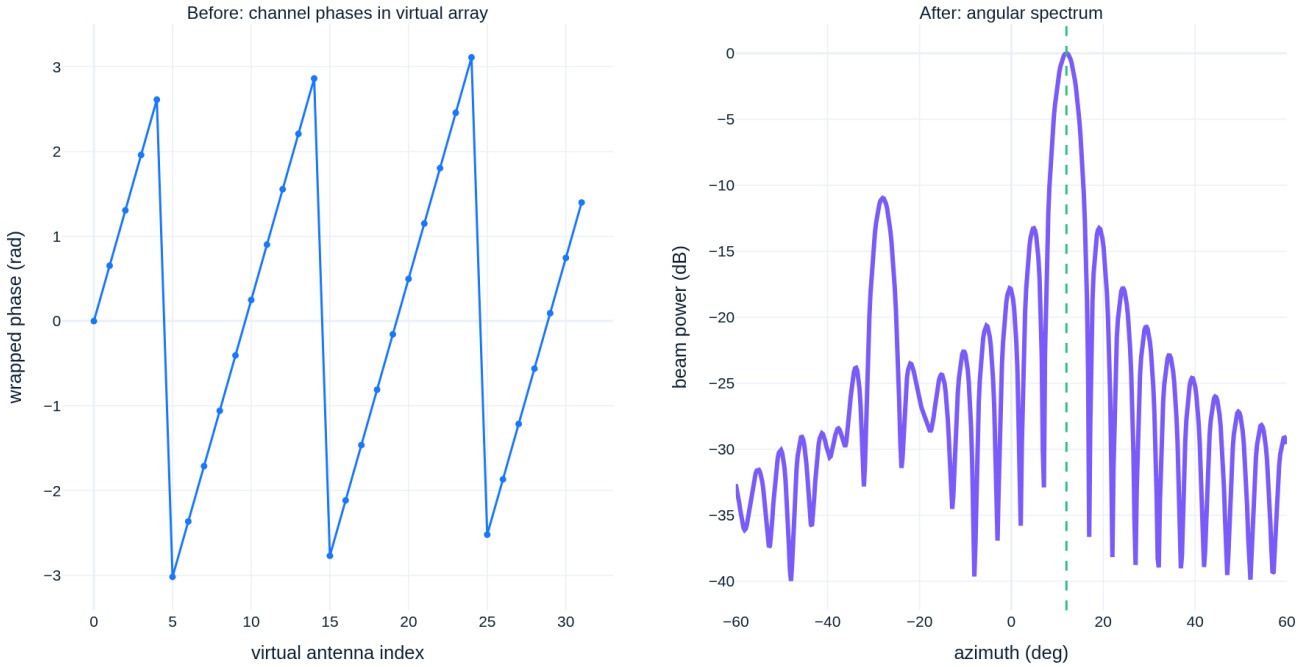




A full 1024 x 256 x 32 complex tensor is around 64 MB at 8 bytes per complex value. The sensor cannot transport every cell. It keeps selected cells and stores bin indices, energy, and the multi-channel beamvector. In the documented interface, 35 records fit in a 1472-byte UDP datagram, with a 5,016-record cap in the failure analysis.

This stage changes the data model from an array to a variable-length list. Ordering, truncation policy, record count, and priority ranking become as important as signal accuracy. A rainy scene can overflow even when every retained record is locally correct.

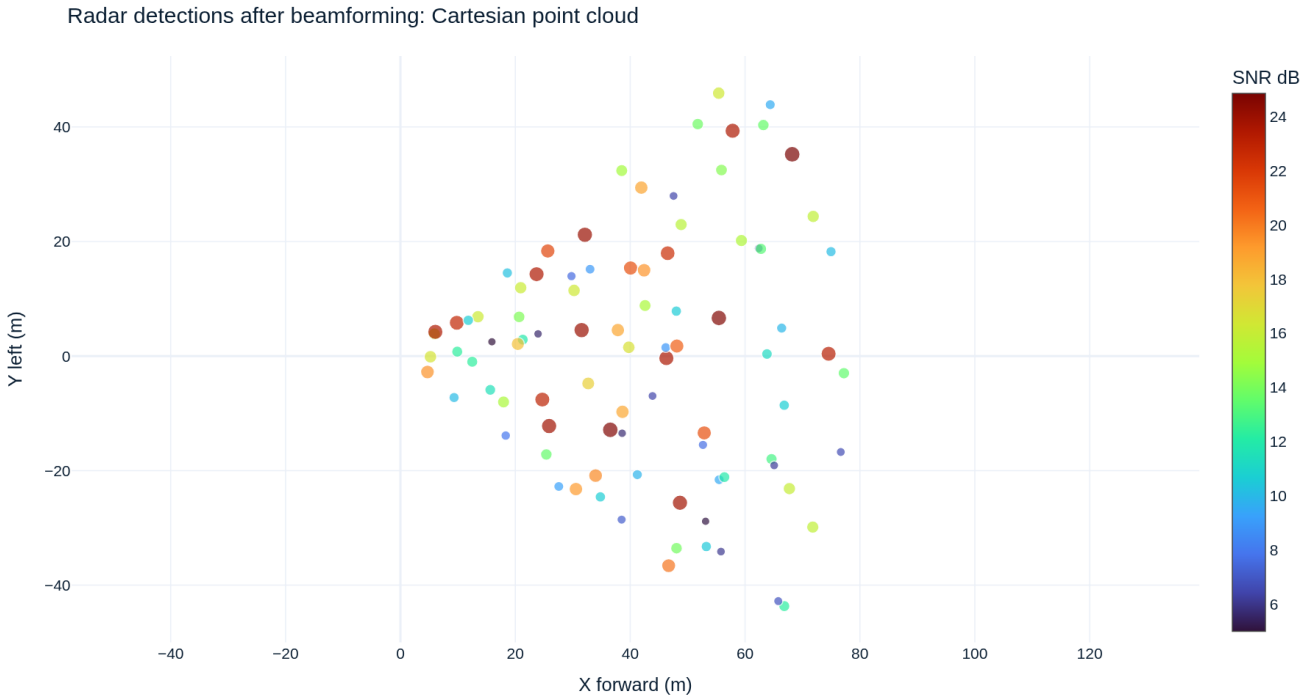
10. Beamforming: channel phase becomes angle



Each virtual antenna sees a phase offset proportional to its position and the target direction. Steering the array over candidate angles and summing coherently produces an angular spectrum. The largest peak estimates azimuth; a second aperture dimension estimates elevation.

Virtual MIMO combines transmitter and receiver positions into a sum co-array. Four transmitters and eight receivers can create 32 virtual channels. Calibration must correct assembly tilt, channel imbalance, and fascia phase delay before angle estimates are trusted.

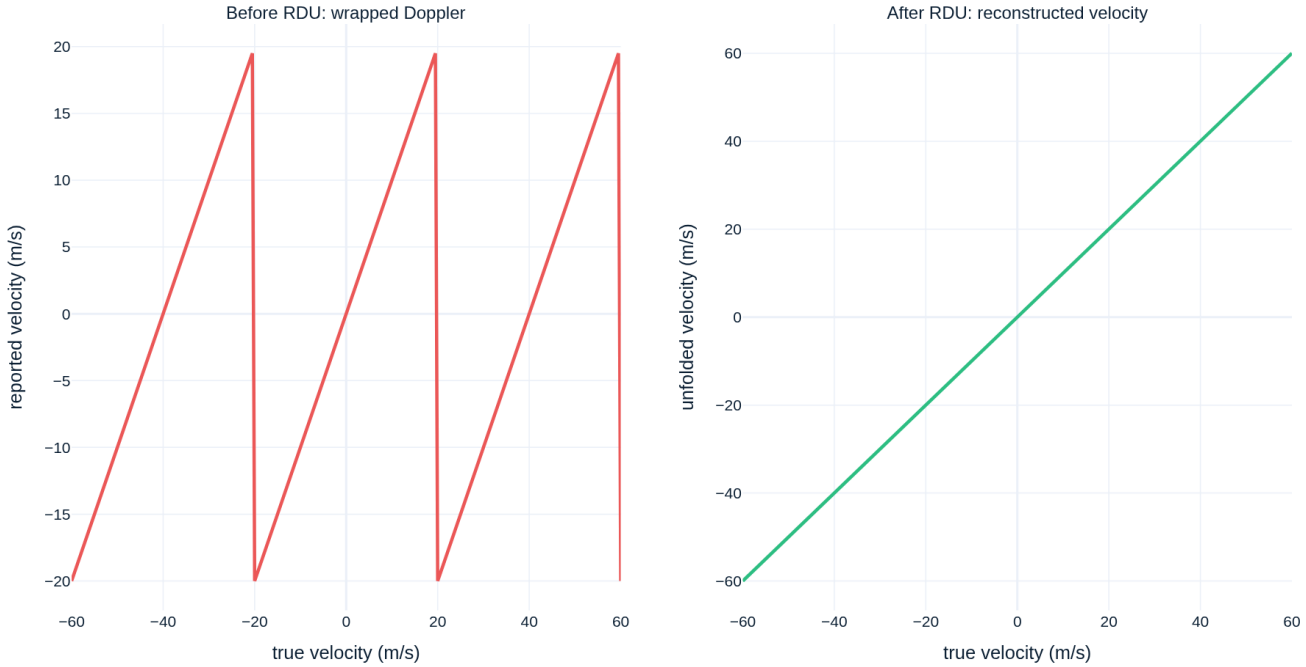
11. Detection list: spectral cells become physical observations



After sub-bin refinement, beamforming, calibration, and metadata attachment, each record becomes a detection such as [range, radial velocity, azimuth, elevation, RCS, SNR, flags]. Converting to Cartesian coordinates gives x=r cos(theta), y=r sin(theta) in the sensor frame.

This is still not an object track. Multiple points may belong to one object, a ghost, a guardrail, rain, or noise. The timestamp and coordinate frame must accompany every detection or multi-sensor fusion will be geometrically inconsistent.

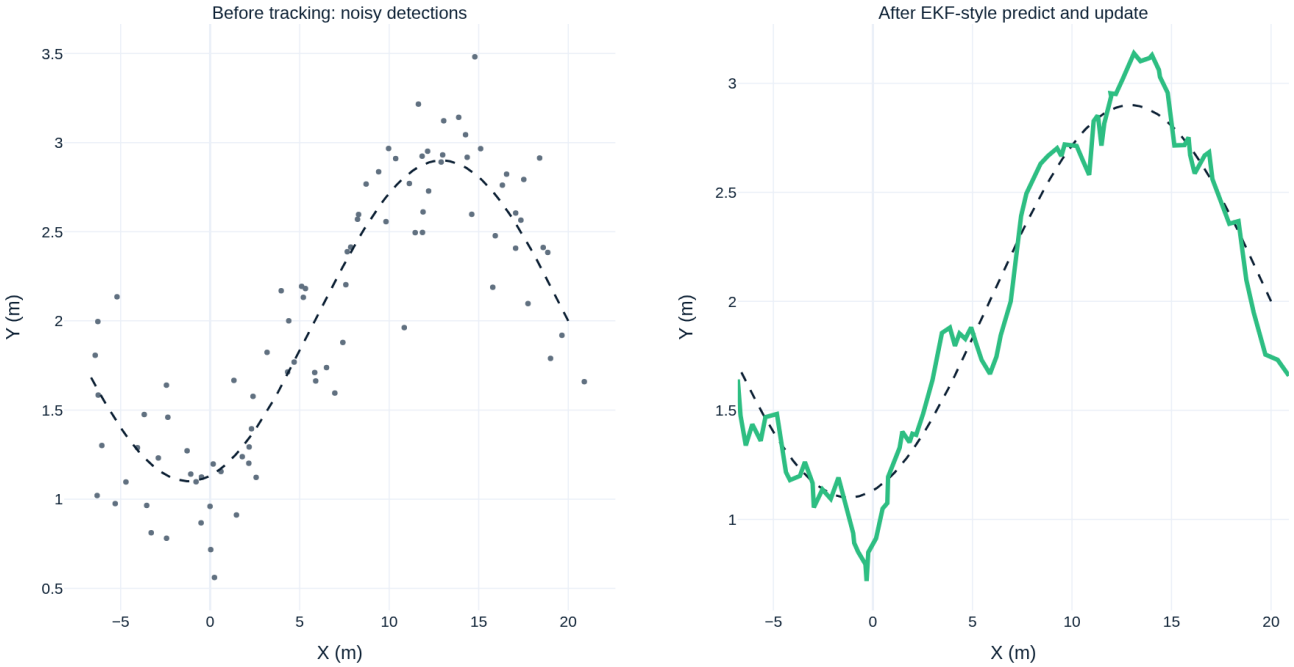
12. RDU: wrapped Doppler becomes physical velocity



Doppler FFT indices are periodic. A fast target can appear at the wrong signed velocity. Range-Doppler unfolding combines staggered pulse-repetition sequences or multiple hypotheses so residues identify a velocity over a wider interval.

A useful RDU test sweeps true velocity through several wrap boundaries and checks continuity after unfolding. It must include opposite sign conventions, low-SNR ambiguity, inconsistent residues, and a refusal path when the data cannot select one hypothesis.

13. EKF tracking: detections become continuous state

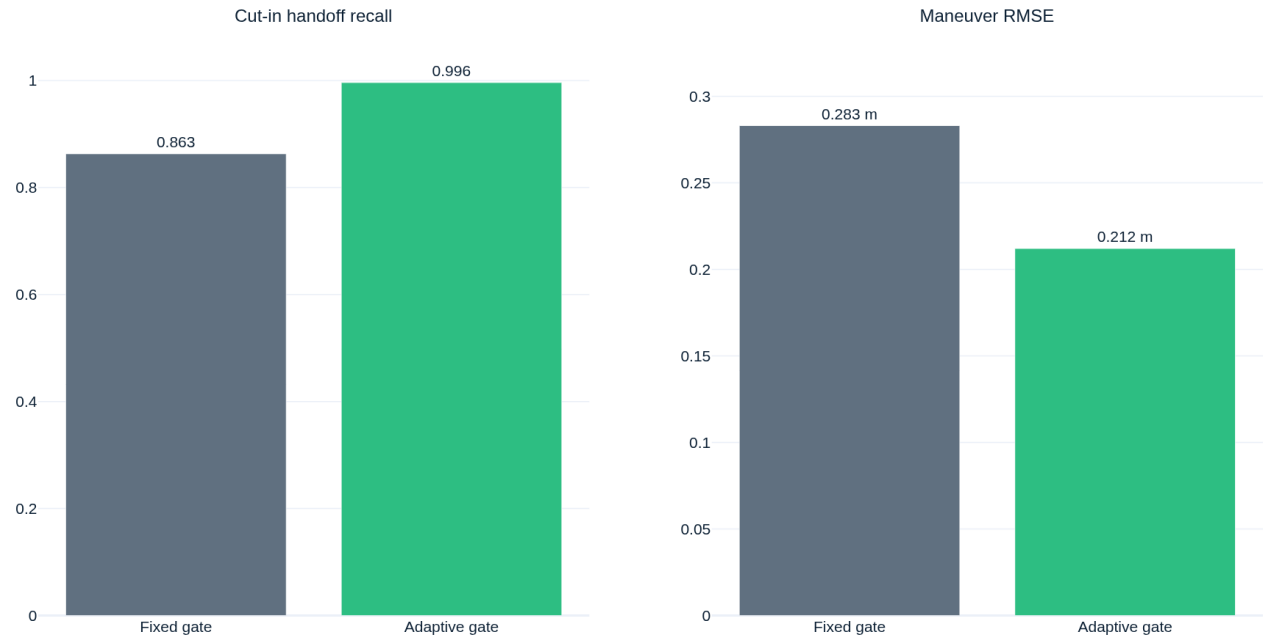


The tracker predicts a state such as [x,y,vx,vy] using a motion model and then updates it using radar measurements. The innovation compares the detection with the predicted measurement. The covariance determines how much the filter trusts prediction versus measurement.

Use the Joseph covariance update P=(I-KH)P(I-KH)^T+KRK^T. It is more robust to fixed-point and finite-precision asymmetry than the shorthand subtraction form.

Association is discrete. Gating, nearest-neighbor or global assignment, birth confirmation, coasting, and deletion decide which detection updates which track. A numerically accurate filter can still fail if association selects the wrong branch.

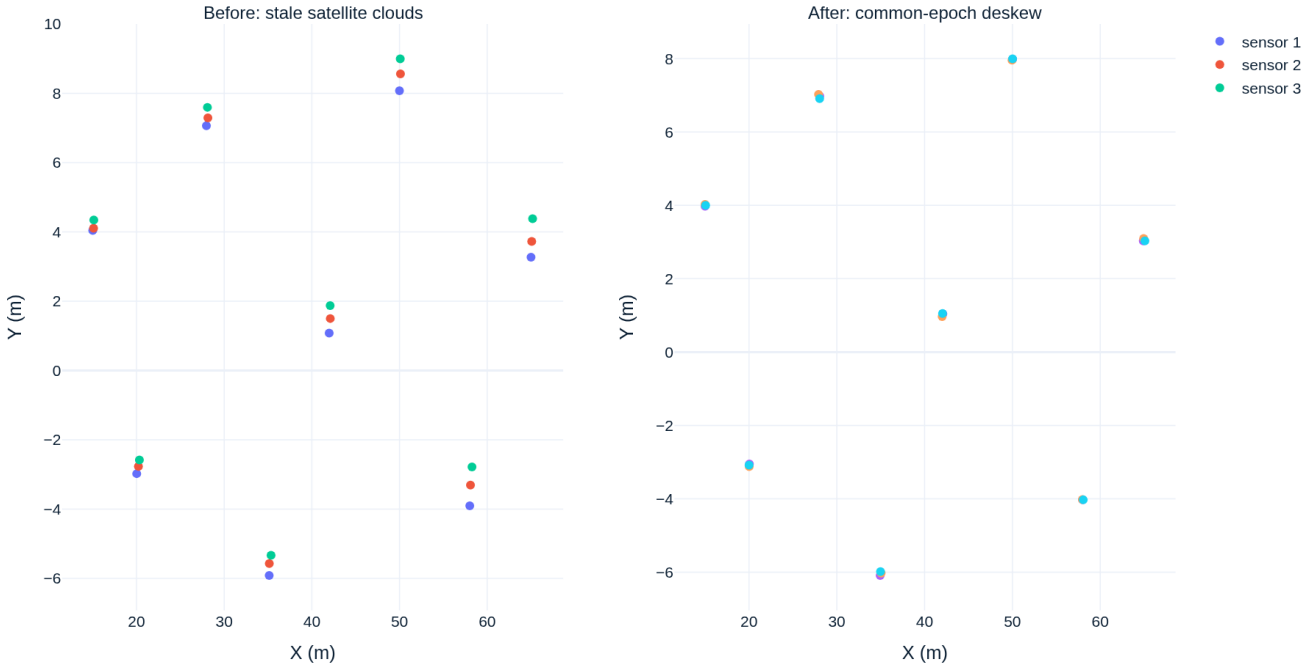
14. Gating before and after: documented cut-in experiment



A fixed chi-square gate lost abrupt cut-in measurements. The bounded adaptive gate used acceleration and innovation energy to expand only during maneuver evidence. In the executed experiment, handoff recall improved from 0.863 to 0.996 and maneuver RMSE improved from 0.283 m to 0.212 m.

The failed IMM-only variant exposed an onset chicken-and-egg problem: the CA-model probability could not rise before the first association, while the first association required a wider gate. This is a code-architecture lesson: the signal that opens a gate must be available before the gate decision.

15. Multi-radar time compensation and fusion



Corner and front radars report in different sensor frames and at different timestamps. Each point must be transformed into the vehicle frame and transported to a common domain-controller epoch using ego rotation and translation. Only then should clouds or tracks be associated and fused.

A representative compensation is p_comp=R_z(omega Delta t)p_meas-v_ego Delta t-0.5 a_ego Delta t^2. The sign of rotation is load-bearing. A sign flip can improve low-rate cases by accident but fail badly at larger yaw rates.

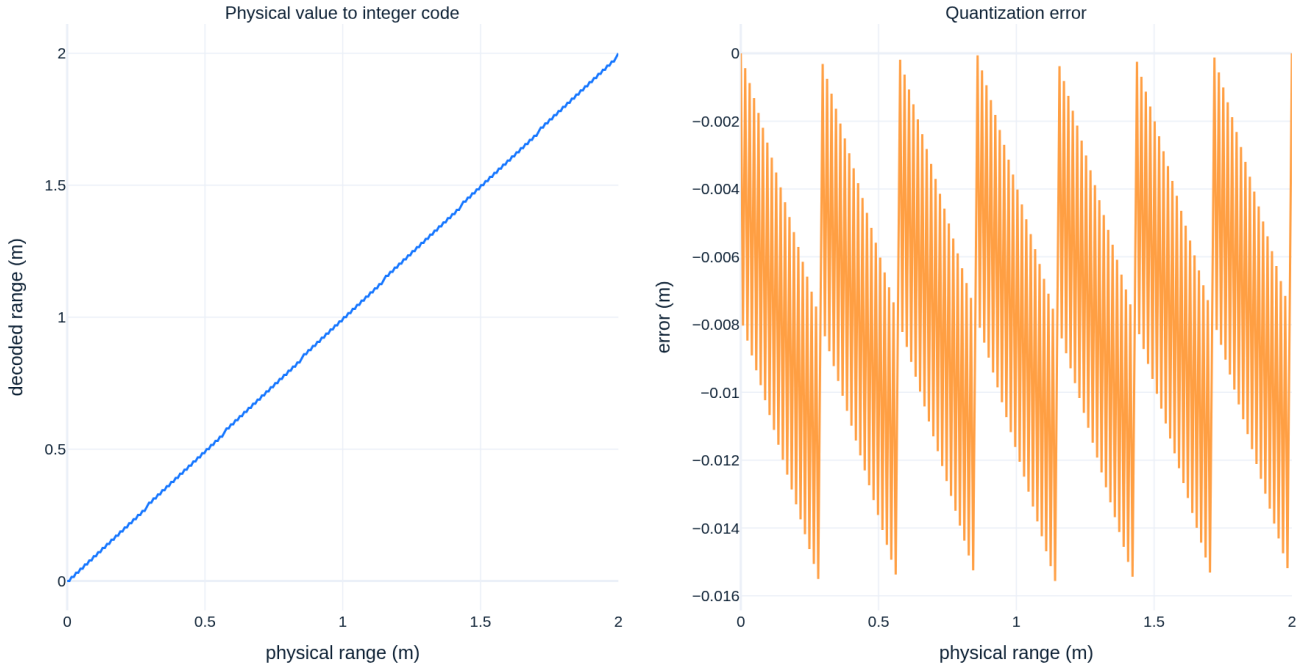
16. Deskew before and after: executed yaw-rate sweep



Across yaw rates 0.1 to 0.6 rad/s, the documented first-order compensation reduced mean fused RMS error by 32% to 86%. Residual error increased at high yaw because second-order rotation, lever-arm effects, and jerk were not fully modeled.

Input contract: sensor timestamp, target position, sensor extrinsic, vehicle velocity, yaw rate, and chosen fusion epoch. Output contract: point or track in a common frame at a common time, plus propagated uncertainty.

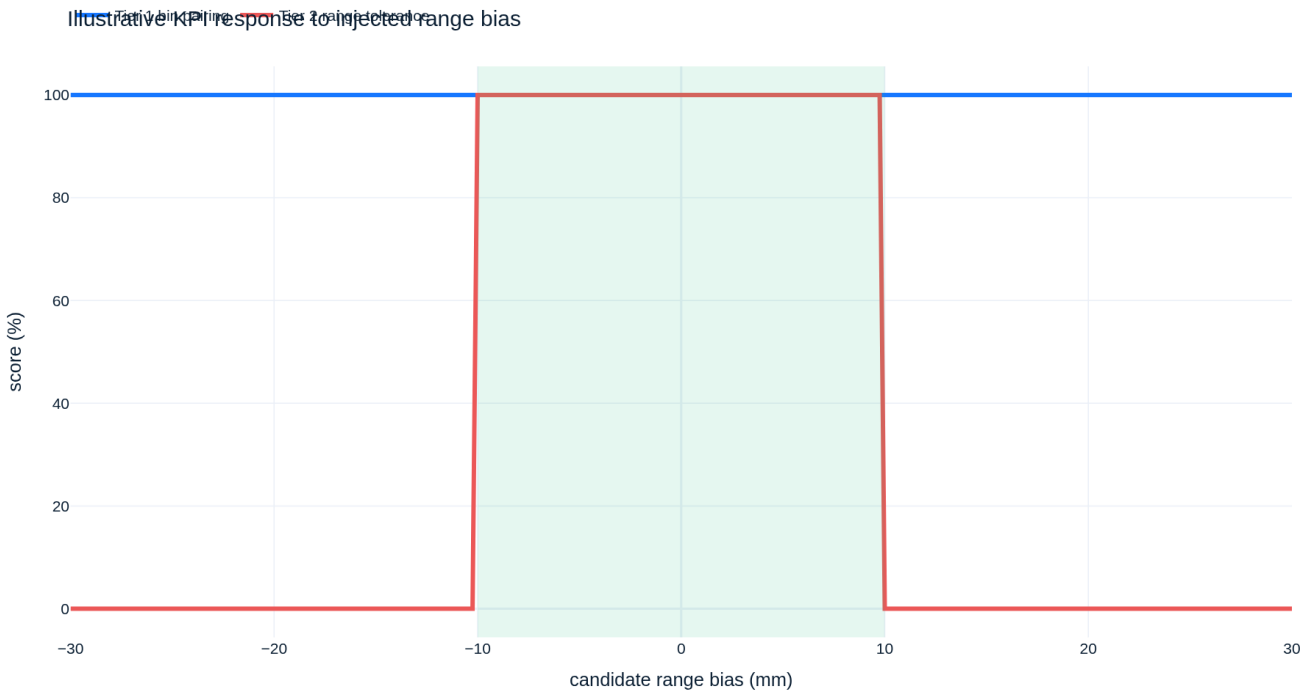
17. CAN-FD serialization: floating physics becomes integers



A DBC field maps physical value to an integer using Raw=floor((Physical-Offset)/Factor). For a 0.015625 m range factor, quantization contributes roughly step/sqrt(12)=4.5 mm RMS under a uniform-error model.

Serialization tests must cover saturation, signedness, endianness, invalid codes, cycle time, message counters, and round trip encode-decode. Matching floating values before serialization does not guarantee matching bus bytes.

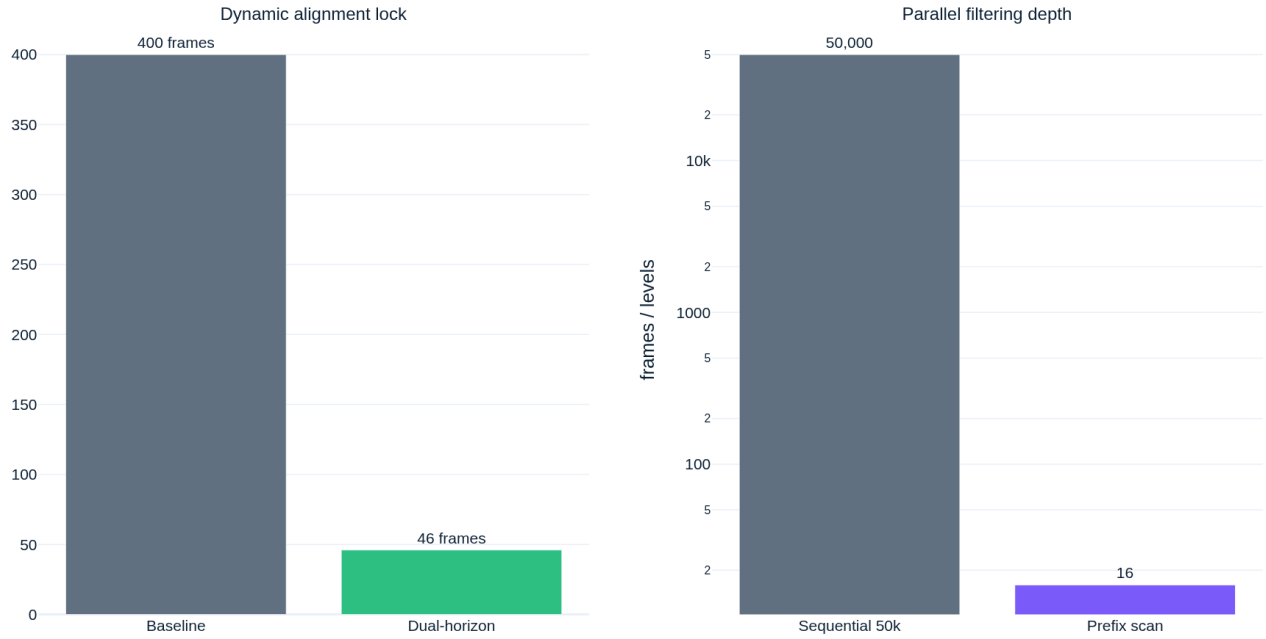
18. KPI validation: compare candidate against reference



Tier 1 asks whether energy or detections pair at the intended discrete indices. Tier 2 evaluates physical-value tolerances after pairing. This is why a +20 mm range-bias candidate can retain 100% Tier-1 yield while achieving 0% Tier-2 accuracy in the documented CSV path.

Layer	Question	Example failure
Tier 1	Did records pair without replacement?	bin shift, truncation, dropped records
Tier 2	Are physical differences within tolerance?	20 mm bias exceeds 10 mm range tolerance
Latency	Did output arrive on time?	more than 40 ms is a regression
CAN matching	Are quantized objects consistent?	code boundary changes despite small float error

19. Replay and HPCC architecture



Sharding a log accelerates replay, but each shard can restart dynamic alignment cold. The documented dual-horizon alignment reduced lock from 400 to 46 frames. Separately, an associative prefix scan reduced theoretical filtering span for 50,000 steps from 50,000 to 16 levels while matching the sequential reference in float64.

These solve different bottlenecks. Fast alignment reduces discarded warm-up data. Prefix scanning reduces dependency depth for a compatible filter formulation. Neither removes I/O, decoding, scheduling, or checkpoint-provenance costs.

20. How the research modules map to the system

Pipeline location	Research modules	Primary question
ADC / beat	n23_residual_track.py	Can an acquired track update range with three products?
Range-Doppler / detection	dual_loop_cfar.py, occupancy_blue.py	Can clutter-aware priors preserve weak targets and false-alarm control?
Ghost geometry	doppler_disambiguation.py, hermite_clothoid.py, specular_contact.py	Can multipath become geometric information rather than only clutter?
Tracking	adaptive_gating.py, spectral_covariance_scaling.py, n24_rank_gate.py	How should uncertainty change under maneuvers, spray, and contamination?
Calibration / fusion	async_motion_compensation.py, n25_ghost_yaw.py, yawfree_clock.py	Can time and angle biases be isolated and certified?
Replay / HPCC	parallel_kf.py, gram_lift.py, guarded_replay.py	Can repeated replay be parallelized without changing decisions?
Synthetic data	doppler_radarsplat_mvp.py	Can range, Doppler, power, and CDC be rendered differentiably?
Validation	hdf5_to_kpi_csv.py, candidate_mf4.py	Can the candidate re-enter KPI tools without production SiL?

21. Recommended code pattern for every new experiment

1. core_algorithm(inputs, parameters) -> outputs
No file I/O, plotting, hidden global state, or unbounded allocation. Keep the mechanism directly testable.

2. fixture(seed) -> inputs, truth
Create a deterministic nominal scenario and explicit counterexamples. If synthetic data is used, label it and retain the generation parameters.

3. metrics(outputs, truth) -> dictionary
Compute recall, precision, RMSE, coverage, false alarms, runtime, memory, and refusal rate as appropriate. Avoid a single metric that can be gamed.

4. demo() and regression tests
Assert equations, units, exact fixtures, numerical tolerances, resource counters, and known failure vectors. Print enough evidence for the ledger.

5. adapter layer
Only after the mechanism passes, connect HDF/MF4 decoding and candidate export. Do not bury algorithm logic in the decoder.

6. storage ledger
Record hypothesis, touched files, raw metrics, anomalies, fixes, and verdict: advance, narrow, discard, or refute. Negative results remain valuable when the mechanism is explained.

22. Worked dummy example: one target from chirp to KPI

Stage	Illustrative value	Interpretation
Scene	target at 60 m, closing 10 m/s	one moving reflector
Chirp	B=1 GHz, T=40 us	S=2.5e13 Hz/s
Mixer	f_b about 10 MHz	range-coded beat
Range FFT	strong peak near expected bin	coarse range
Doppler FFT	negative Doppler bin	closing target
CFAR	peak exceeds adaptive threshold	candidate cell
CDC	bin, energy, 32-channel beamvector	sparse transport record
Beamforming	azimuth peak at 12 deg	direction estimate
Detection	r, v_r, az, el, SNR	physical observation
RDU	select unwrapped -10 m/s	velocity hypothesis
EKF	update x,y,vx,vy and P	smoothed track
Fusion	deskew to t_DC and VCS	common frame and epoch
CAN	quantize with DBC factors	bus bytes
KPI	Tier-1 pair plus Tier-2 tolerance	regression verdict

Important: the plots in this guide that illustrate waveform shape, point clouds, wraparound, and quantization are synthetic teaching examples. The before-and-after performance numbers explicitly identified as documented results come from the executed research ledger.

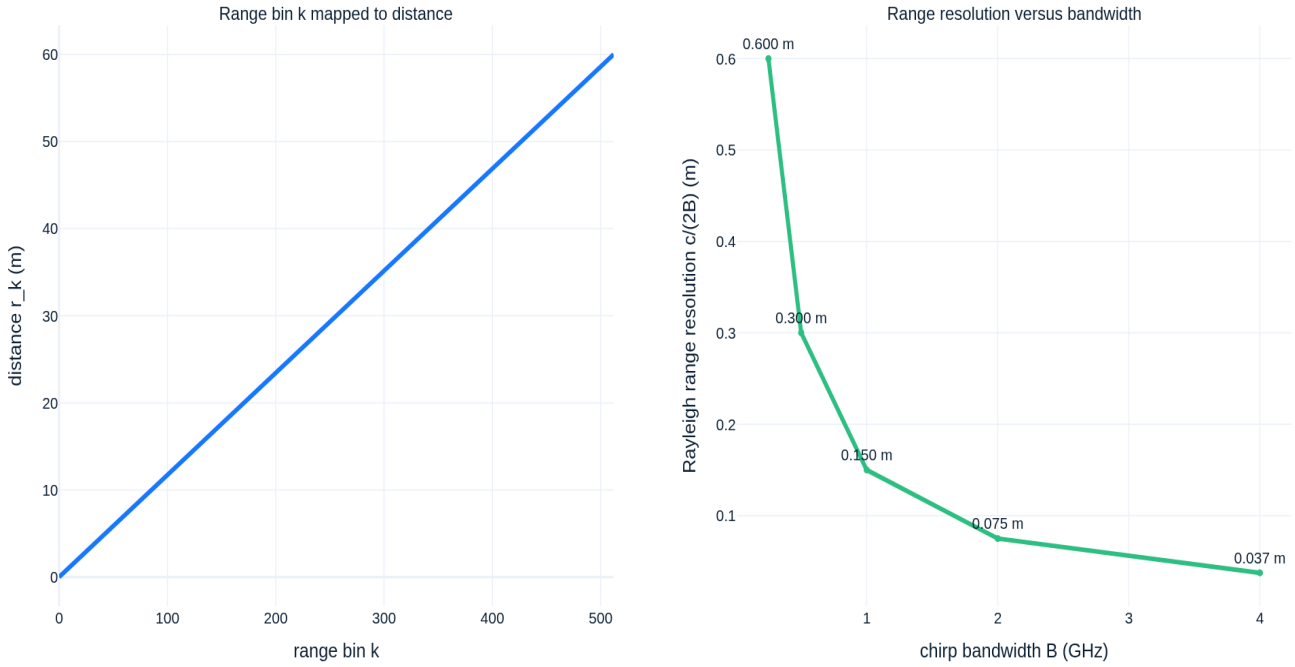
23. Integration checklist

Check	Pass condition
Units	Every interface declares SI units, scale, and sign convention
Shape	Dimensions and axis order are explicit
Time	Timestamp source and target epoch are explicit
Frame	Sensor, vehicle, or world frame is explicit
State	Reset, warm start, and checkpoint behavior are tested
Determinism	Fixed seed produces repeatable metrics
Numerics	Overflow, saturation, NaN, singularity, and strict boundaries are tested
Association	Discrete decisions and alternative hypotheses are covered
Resources	Operation count, memory, heap behavior, and target timing are measured
Validation	At least one exact fixture, one noise ladder, and one kill vector
Evidence	Raw metrics and failures are written to storage ledger
Promotion	Production integration occurs only after a fair baseline comparison

Suggested next extension

Add screenshots or schemas from real HDF frames beside the synthetic diagrams, then link each pipeline stage to the exact dataset path and field names used by your decoder. This will turn the guide from a conceptual architecture manual into an onboarding and debugging handbook for the repository.

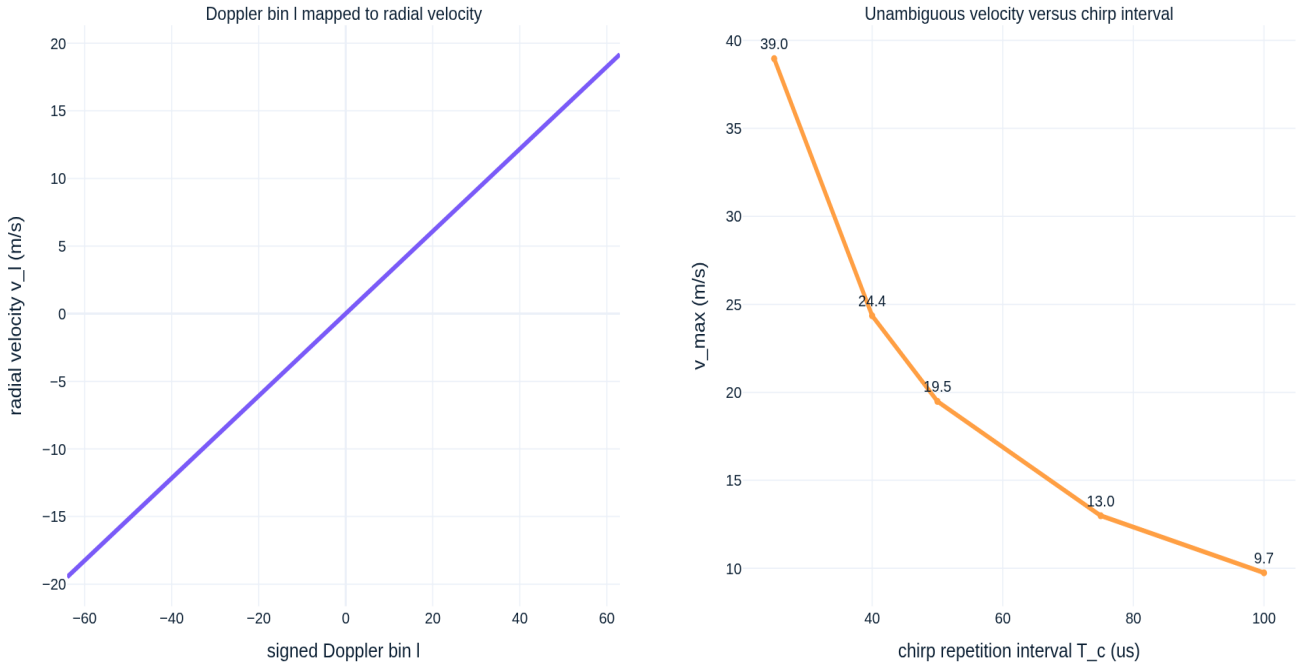
24. Range bin k versus physical distance



A range FFT does not directly output metres. It outputs an integer bin k. Calibration constants convert that bin to distance: r_k = k c /(2 S T_s N_R). The left plot makes this mapping explicit. The right plot separates bin spacing from fundamental range resolution: increasing bandwidth improves the ability to separate nearby targets.

Intuition check: doubling N_R with the same sampled bandwidth gives a denser frequency grid, but it does not automatically double physical resolving power. Bandwidth determines the Rayleigh resolution c/(2B); zero padding mainly interpolates the displayed spectrum.

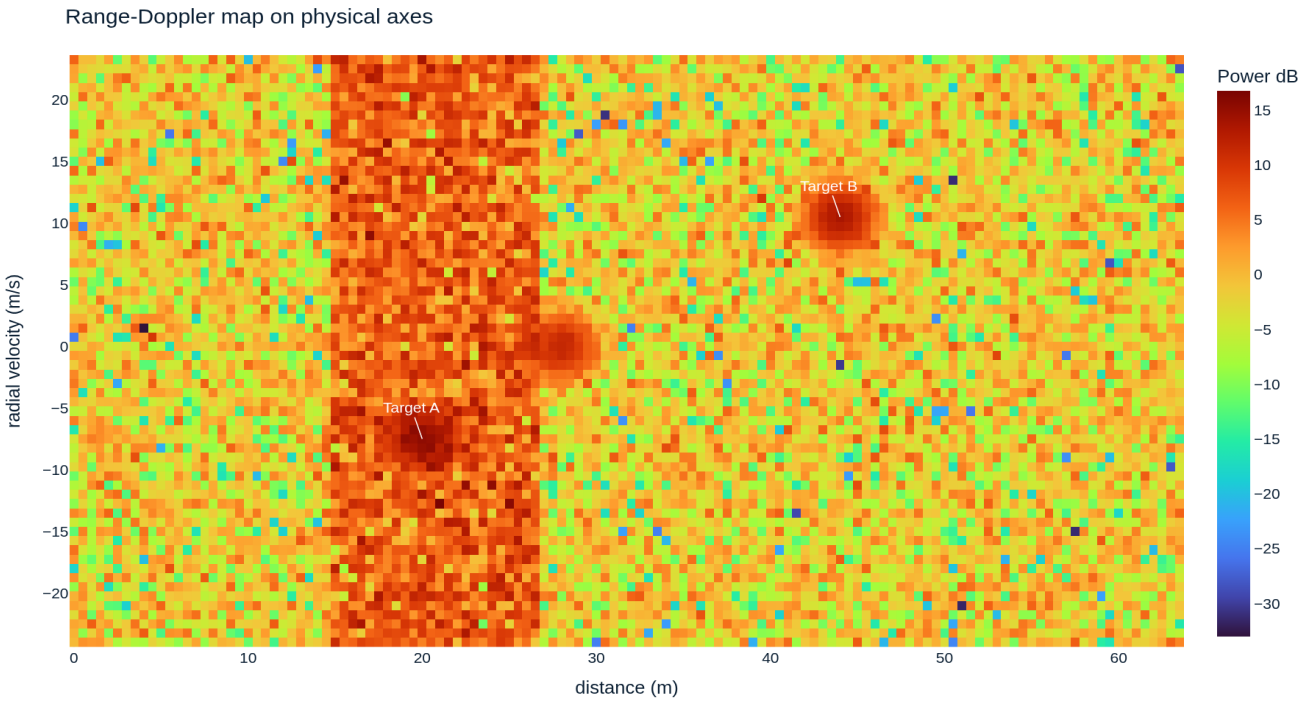
25. Doppler bin l versus radial velocity



The signed Doppler index l maps linearly to radial velocity: v_l=l lambda/(2N_D T_c). Negative and positive bins represent opposite radial directions under the selected sign convention. The index stored in memory may be unsigned, so bins above N_D/2 often represent negative signed bins after fftshift-style interpretation.

The right plot shows the trade-off: shorter chirp intervals increase unambiguous velocity, while more chirps improve Doppler-bin spacing but increase dwell time. A complete design must balance velocity range, resolution, latency, and target acceleration.

26. The most important intuition plot: distance versus velocity

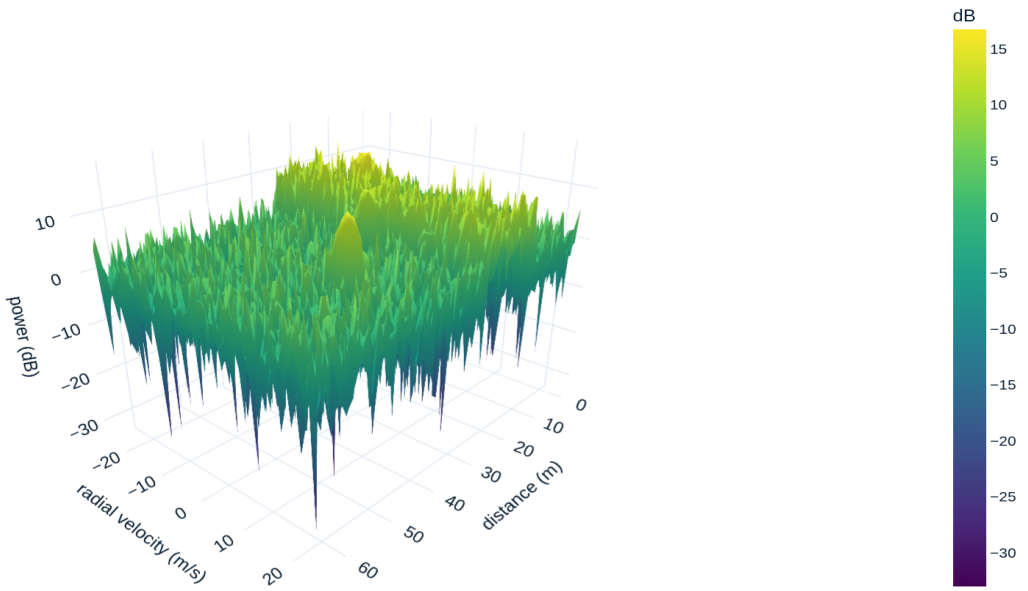


This is the Range-Doppler matrix expressed in physical units rather than abstract k and l indices. Every bright cell answers two questions simultaneously: at what distance is the reflector, and how fast is its radial motion? A vertical structure indicates energy over many velocities at similar range; a horizontal structure indicates energy across many ranges at similar Doppler.

The conversion is performed with two independent rulers: k to metres using chirp slope and sample timing, and l to metres per second using wavelength, chirp interval, and chirp count. Angle is not yet present in this 2D map.

27. The same Range-Doppler data as a 3D surface

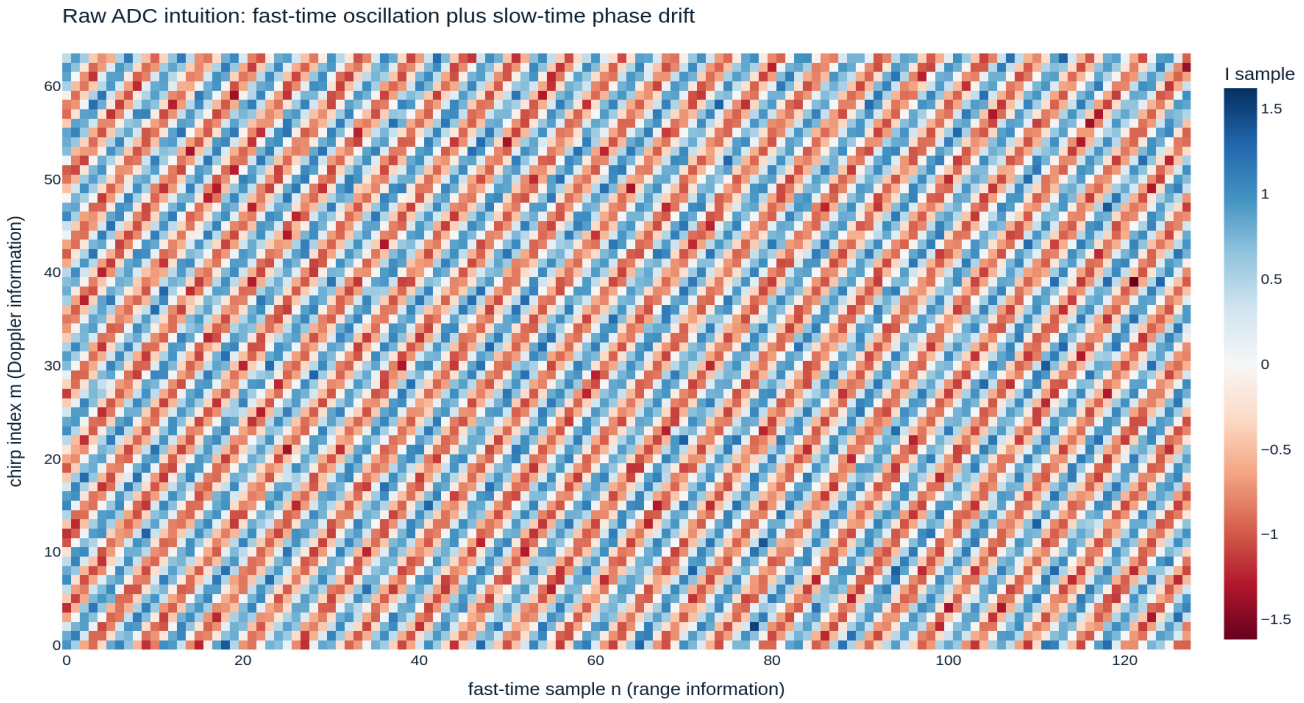
3D view: peaks rise above the clutter floor



A heatmap shows location well; a 3D surface shows peak height relative to clutter. CFAR is conceptually trying to decide whether a local hill is unexpectedly tall compared with its surrounding terrain. Strong guardrail or interference ridges can lift the local terrain and hide a real target.

Use the heatmap for exact bin localization and the 3D surface for intuition. Do not use perspective height alone for quantitative comparison, because camera angle and color scaling can exaggerate or hide differences.

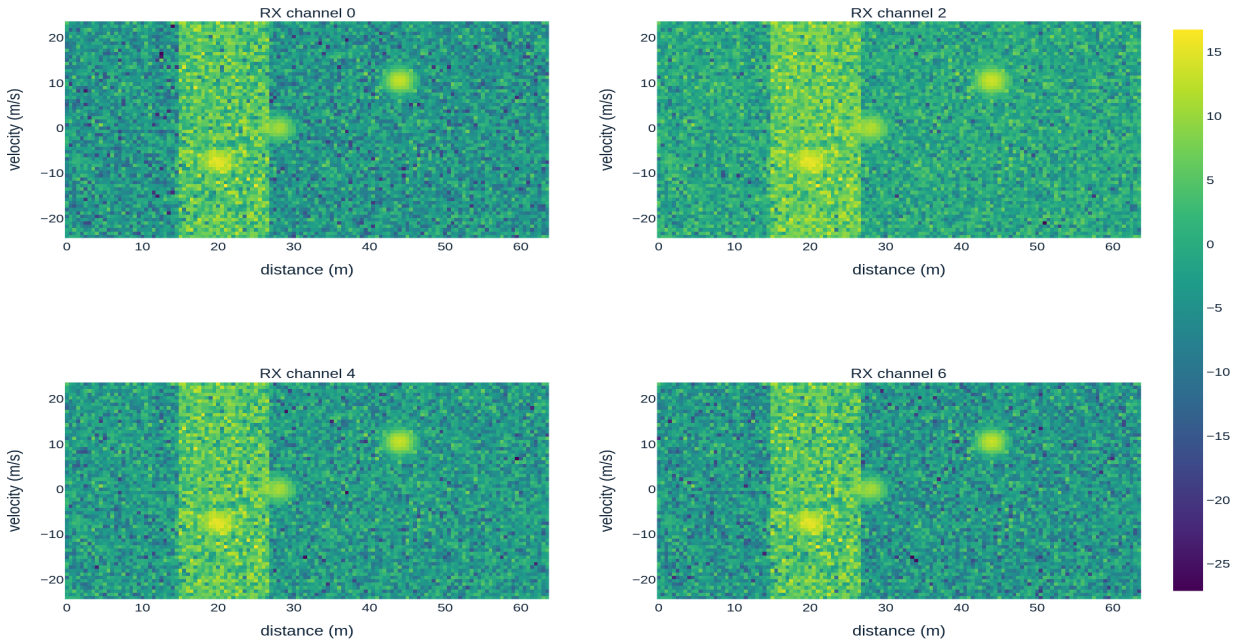
28. Raw fast-time and slow-time axes



Each row is one chirp and each column is one ADC sample inside that chirp. Oscillation along a row carries beat frequency and therefore range. Phase drift from row to row carries Doppler and therefore radial velocity. A 2D FFT simply applies the correct spectral ruler along each of these orthogonal axes.

This figure explains why swapping axes in code is disastrous: the program would interpret chirp-to-chirp motion as range and within-chirp beat pitch as Doppler. Every array adapter should state axis order explicitly.

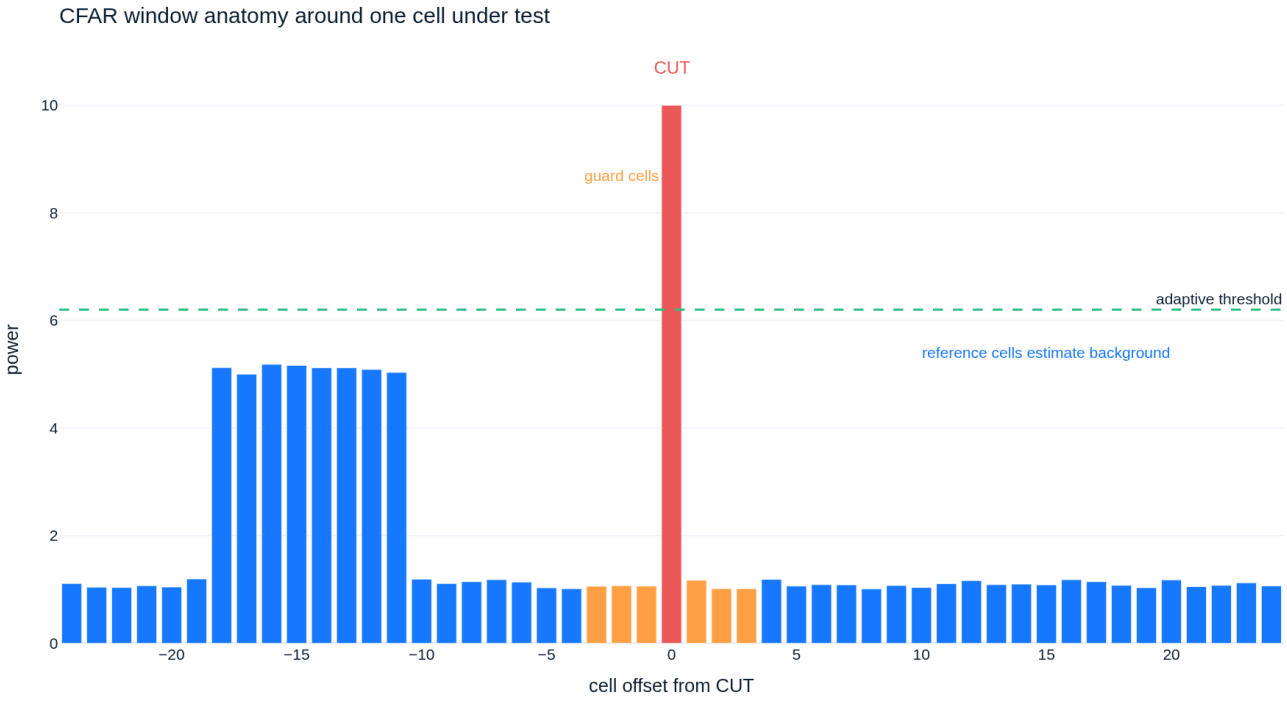
29. Radar data cube: why it is 3D before compression



The Range-Doppler surface exists separately for each receive or virtual antenna channel. The channel-to-channel complex phase differences are later used for angle estimation. Therefore the pre-beamforming cube has at least three important axes: range, Doppler, and antenna channel. Frames and elevation apertures add further dimensions.

A power-only collapse across channels can preserve detection energy while destroying angle information. The CDC therefore retains a beamvector or equivalent channel data for selected cells.

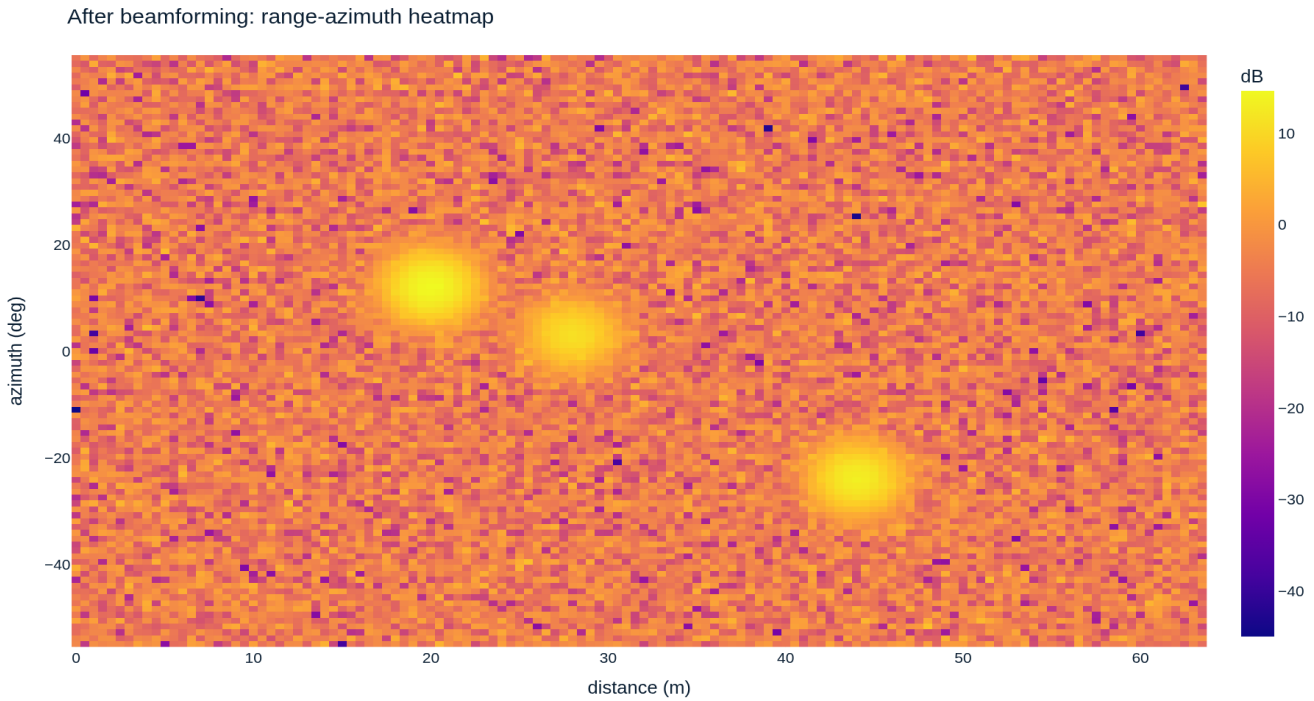
30. CFAR window anatomy



The red cell is the cell under test. Orange guard cells prevent the target main lobe and immediate sidelobes from contaminating the background estimate. Blue reference cells estimate local clutter or noise. The threshold multiplier converts that estimate into the requested false-alarm operating point under the assumed distribution.

Failure modes to visualize: a wall occupying one side of the window, two nearby targets contaminating reference cells, censored cells changing effective N, and a target spanning more cells than the guard region. The dual-loop and Occupancy-BLUE work address different parts of these failures.

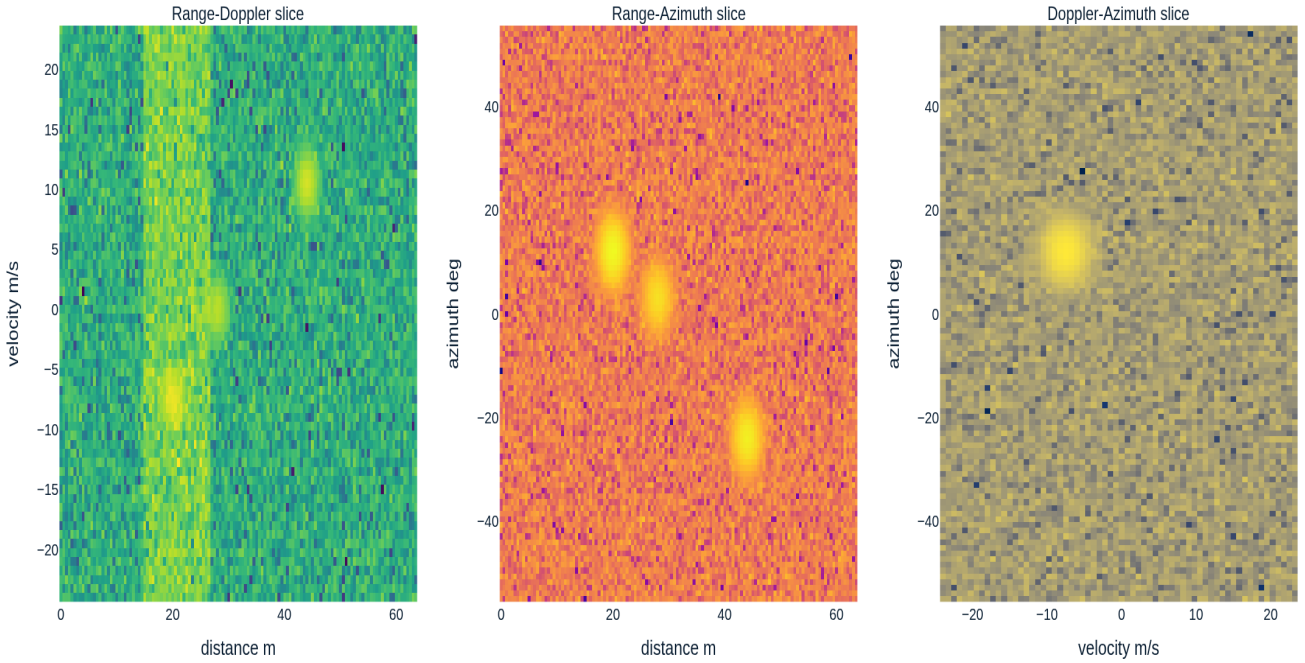
31. Range-Azimuth map after beamforming



After angle processing, detections can be viewed on a distance-versus-azimuth grid. This separates two targets at the same range but different directions, something a Range-Doppler slice alone may not do. The conversion to Cartesian position then uses x=r cos(theta) and y=r sin(theta).

Range-Azimuth is not the same as a camera image. Angular bins widen in metres with distance: one degree corresponds to about 0.17 m laterally at 10 m but 1.75 m at 100 m. This is why positional uncertainty must be range dependent.

32. Range-Doppler-angle as linked slices

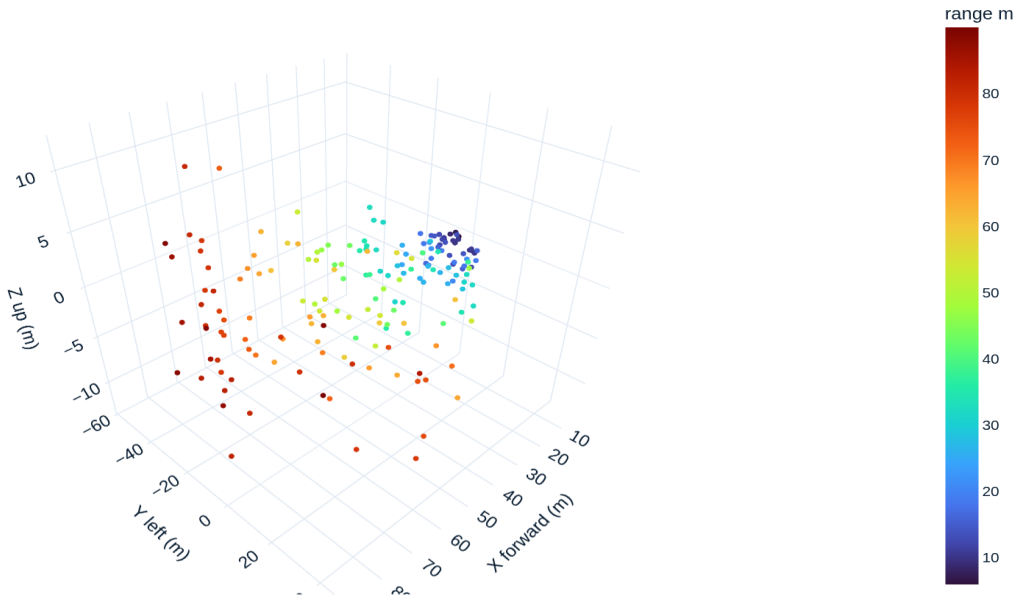


A practical radar tensor is difficult to show in one static figure. Three linked projections are more useful: Range-Doppler identifies distance and radial speed; Range-Azimuth identifies distance and direction; Doppler-Azimuth identifies moving directions. A selected cell in one view should map consistently into the other views.

Debugging method: select one known synthetic target and mark its expected location on all three projections. If only one projection is wrong, the fault is likely in that axis transform rather than the RF scene generation.

33. 3D detections after elevation estimation

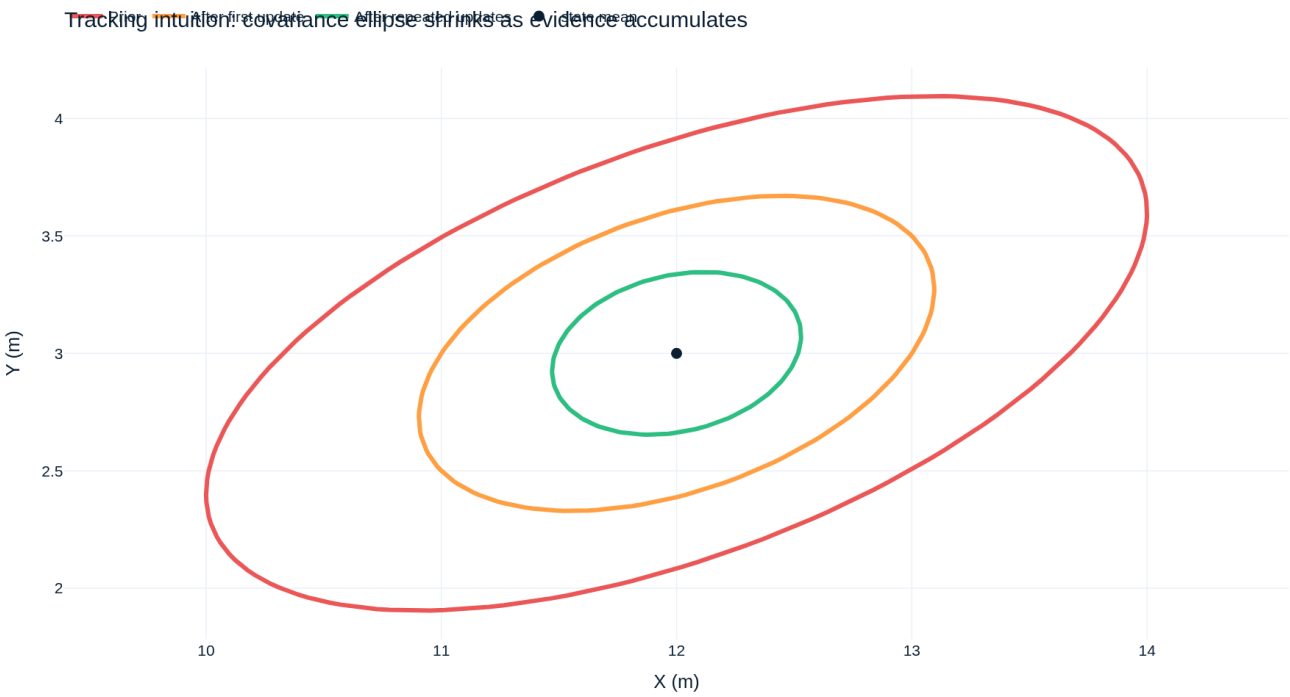
3D radar detection cloud after azimuth and elevation estimation



Once azimuth and elevation are available, each detection maps to X, Y, and Z. This visualization is useful for overpasses, trucks, road surfaces, and vertical multipath. Automotive radar elevation resolution is usually coarser than azimuth, so Z uncertainty may be much larger than the apparent marker cloud suggests.

Always keep range, angle, and covariance with the Cartesian point. Converting to XYZ does not make uncertainty isotropic. Angle noise creates cross-range error that grows with distance.

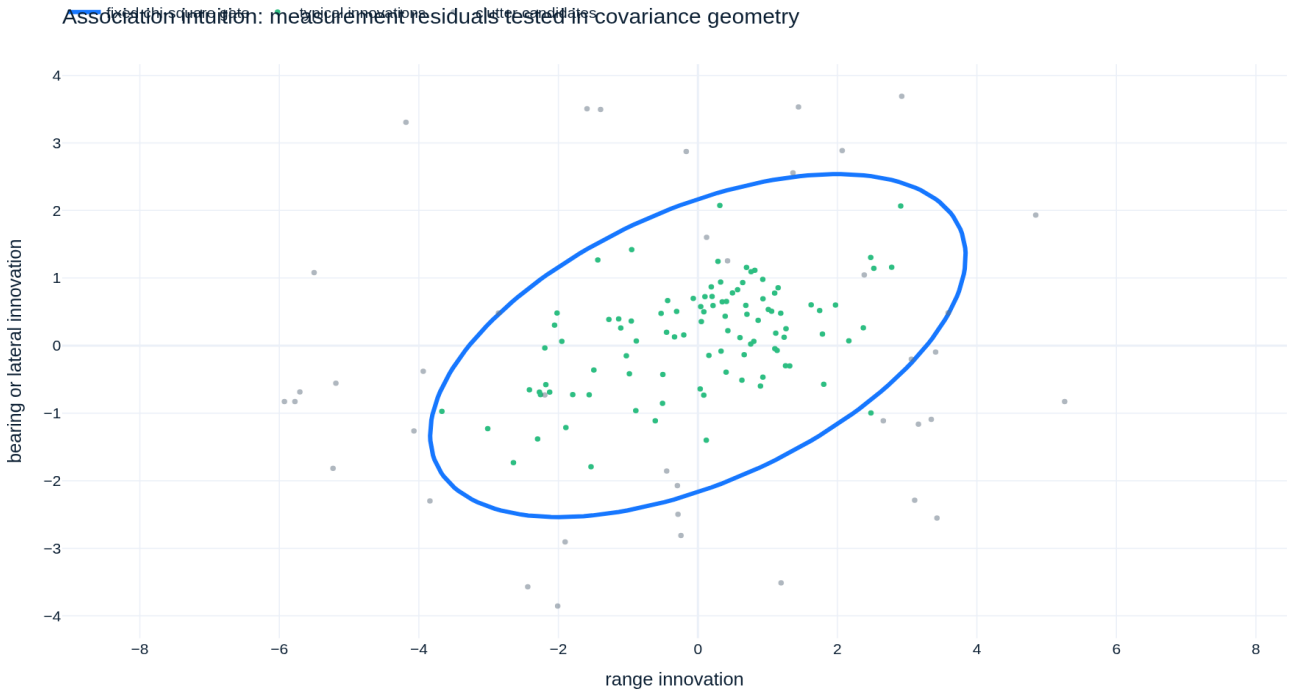
34. Tracking covariance is part of the data



A track is not only a mean position. It is a probability or uncertainty region represented by covariance. Prediction generally enlarges the ellipse; informative measurements shrink it. The ellipse orientation records correlated uncertainty, which affects gating, fusion, and downstream confidence.

Do not compare only track centres. A filter producing a small error with overconfident covariance can be more dangerous than one producing slightly larger error with honest uncertainty. Report RMSE together with consistency or coverage.

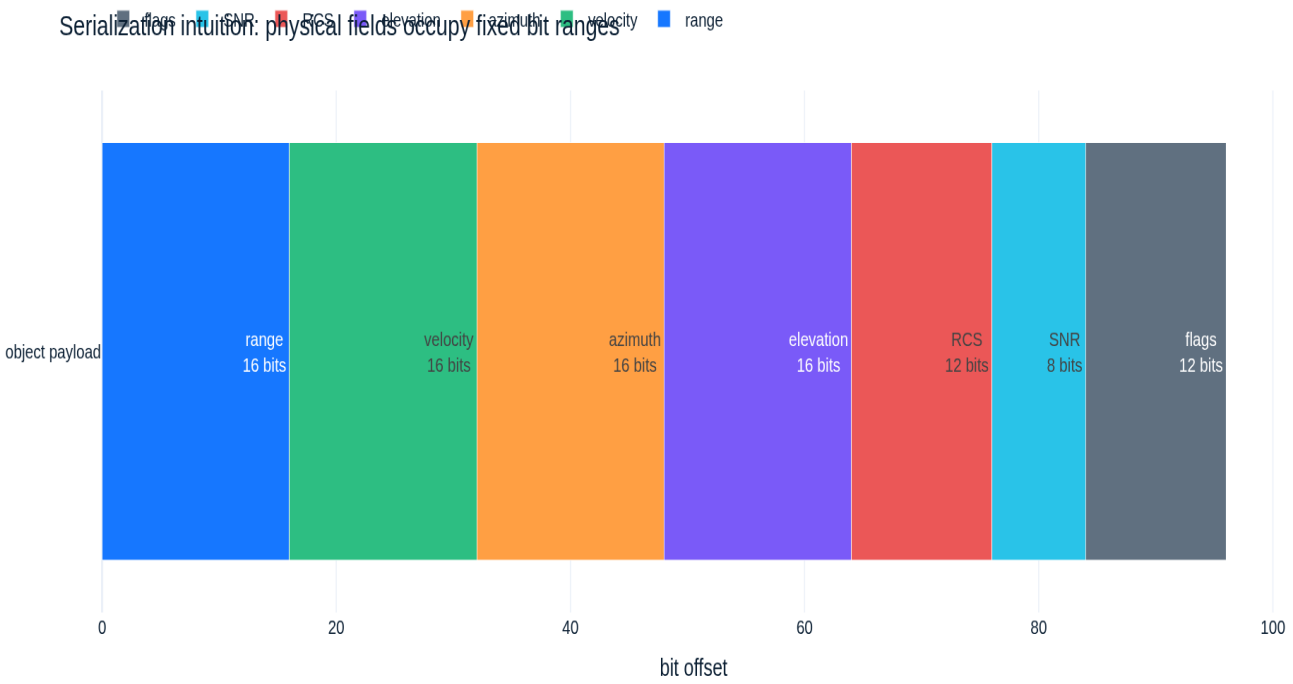
35. Association gate in innovation space



The gate is an ellipse because innovations are normalized by covariance. The same raw residual can be acceptable in a high-uncertainty direction and unacceptable in a precise direction. Mahalanobis distance measures this geometry.

Adaptive gating changes the admissible region, but it must remain bounded. If it expands whenever innovations become large, clutter can force the gate open and create a positive feedback loop. The cut-in experiment therefore measured both recall and RMSE.

36. CAN payload as a bit-level data structure



After floating-point or fixed-point perception, the bus contract allocates exact bit ranges to each value. A change in factor, offset, signedness, endianness, or invalid code changes the interpreted physical value without changing the upstream algorithm.

A robust validation stack checks three levels separately: numerical state before serialization, encoded bytes, and decoded physical values. This isolates algorithm regressions from interface regressions.

37. Complete axis and unit checklist

Representation	Axis 1	Axis 2	Axis 3 / metadata
Raw ADC	sample n: seconds	chirp m: count	RX channel, complex I/Q
Range spectrum	range bin k	chirp m	RX channel, complex phase
Range-Doppler	range bin k	Doppler bin l	RX channel or power
Range-Azimuth	distance m	azimuth deg	power / SNR
Detection list	record index	physical fields	timestamp, sensor frame
Track list	track ID	state dimensions	covariance, age, status
Fusion list	global track ID	VCS state	common epoch, source mask
CAN	message ID	byte / bit offset	factor, offset, invalid code
KPI	reference-candidate pair	metric dimension	tolerance and latency

If a plot is missing a unit, sign convention, axis order, timestamp, or coordinate frame, it is not yet a reliable engineering plot. These labels should be generated from the same configuration used by the algorithm whenever possible.

38. Source and evidence note

Primary internal sources used: radar_monograph.pdf for the end-to-end FMCW, DSP, tracking, fusion, KPI, and research explanations; autoresearch_research.ideas.md for research-node mechanics, hardware constraints, and failure scopes; storage.md for executed metrics, code touchpoints, fixes, and verdicts.

Documented result highlights reproduced here include dual-loop CFAR recall 0.50 to 1.00 with false alarms 1 to 0; adaptive gate recall 0.863 to 0.996 and RMSE 0.283 m to 0.212 m; async compensation error reduction of 32% to 86%; 400-frame alignment reduced to 46 frames; and 50,000-step sequential dependency reduced to 16 scan levels for the verified associative formulation.

This guide does not claim that all synthetic plots are measurements from production radar. They are explanatory visuals generated from stated dummy parameters. Patentability, production readiness, and target-hardware timing remain separate evidence questions.